Analysis 1: Aug-Nov 2026

(I) Let $f : (a, b) \mapsto (c, d)$ and $g : (c, d) \mapsto (u, v)$ be differentiable functions, with derivative f' and g'. Let $h(x) = g(f(x))$ for $x \in (a, b)$. Show that h is also differentiable on (a, b) and its derivative h' is given by

$$h'(x) = g'(f(x))f'(x).$$

(II) Let $E : \mathbb{R} \mapsto (0, \infty)$, $F : (-1, 1) \mapsto \mathbb{R}$ and $G : (-1, 1) \mapsto \mathbb{R}$ be defined by $E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, $x \in \mathbb{R}$, $F(x) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k}$, $x \in (-1, 1)$ and $G(x) = E(F(x))$, $x \in (-1, 1)$.

(a) Show that E, F, G are differentiable on their respective domains, with derivatives E', F', G' and:

$E'(x) = E(x)$, $\forall x \in \mathbb{R}$,

$F'(x) = \frac{1}{1+x}$,

$G'(x) = \frac{1}{1+x}$.

(b) For $x \in (0, 2)$, let $L(x) = G(x - 1)$ and $U(x) = E(L(x))$. Show that

$U'(x) = 1$ for $x \in (0, 2)$.

$U(x) = x$ for $x \in (0, 2)$.